INCH-POUND
MIL-DTL-24231/10E(SH)
26 April 2013
SUPERSEDING
MIL-C-24231/10D(SH)
w/AMENDMENT 1
2 March 1992
MIL-C-24231/10D(SH)
15 May 1987

DETAIL SPECIFICATION SHEET

CONNECTORS, SINGLE CABLE, TYPE III, 14-, 24-, 30-, OR 40-CONDUCTOR

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24231.

REQUIREMENTS:

1. Dimensions and configuration: see figures 1 through 25.

2. The gland nut shall be tightened until the packing tightens against the wires and cables. This can be determined by moving the wires or cables gently back and forth until the wire or cable is seized by the packing. The gland nut shall be turned approximately one-fourth of a turn more to hold the wires and cables in place. Do not overtighten the gland nut or attempt to tighten down metal to metal, as this can damage the wires and cables.

3. Full or partial threads shall not extend more than 0.083 inch beyond the ¾-inch thread length.

4. Refer to MIL-DTL-24231/5 for alternate method of manufacturing connector body and installing guide pin M24231/5-071.

5. The retainer plates and the retainer plate assemblies shall be manufactured from nickel-copper alloy in accordance with Class A, Form 1, hot-finished condition of QQ-N-281, with a minimum yield strength of 40,000 lbf/in^2 at 0.2 percent offset, or nickel-copper alloy in accordance with Class A, Form 1, cold-drawn condition of QQ-N-281 with a minimum yield strength of 50,000 lbf/in^2 at 0.2 percent offset.

6. For welding the pin, item 35, see MIL-DTL-24231/5.

7. Unless more stringent requirements are specified, retainer plate, packing and retainer plate assembly for other special conductor arrangements (for example, LS2SWU-1 and B-16 wire combinations) shall meet the following minimum requirements:

a. Positional tolerancing for all holes shall be ⌖ | Ø.010 .

b. Holes shall not be located closer than 0.010 from the minimum gland nut inside diameter of part number M24231/10-044 and the retainer plate position pin holes of part number M24231/10-080.

c. Minimum material between holes shall be 0.031.

d. Protective sleeving shall be sized to suit the number of cables required where use of the specified sleeving, item 12 or 13, is inadequate.

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8. All dimensions are in inches.

9. Tolerances shall be as follows, or as stated:

a. Decimals – ±0.005 inch.

b. Fractions – ±1/64 inch.

c. Angular dimensions – ±½ degree.

d. Tolerances for rubber packing should follow the guidance provided in table XLI (RMA Drawing Designation A2 Precision) of MIL-HDBK-149.

10. The flange outside diameter is intended to coincide with commonly available 3½-inch bar stock. The dimension may vary after machine clean-up.

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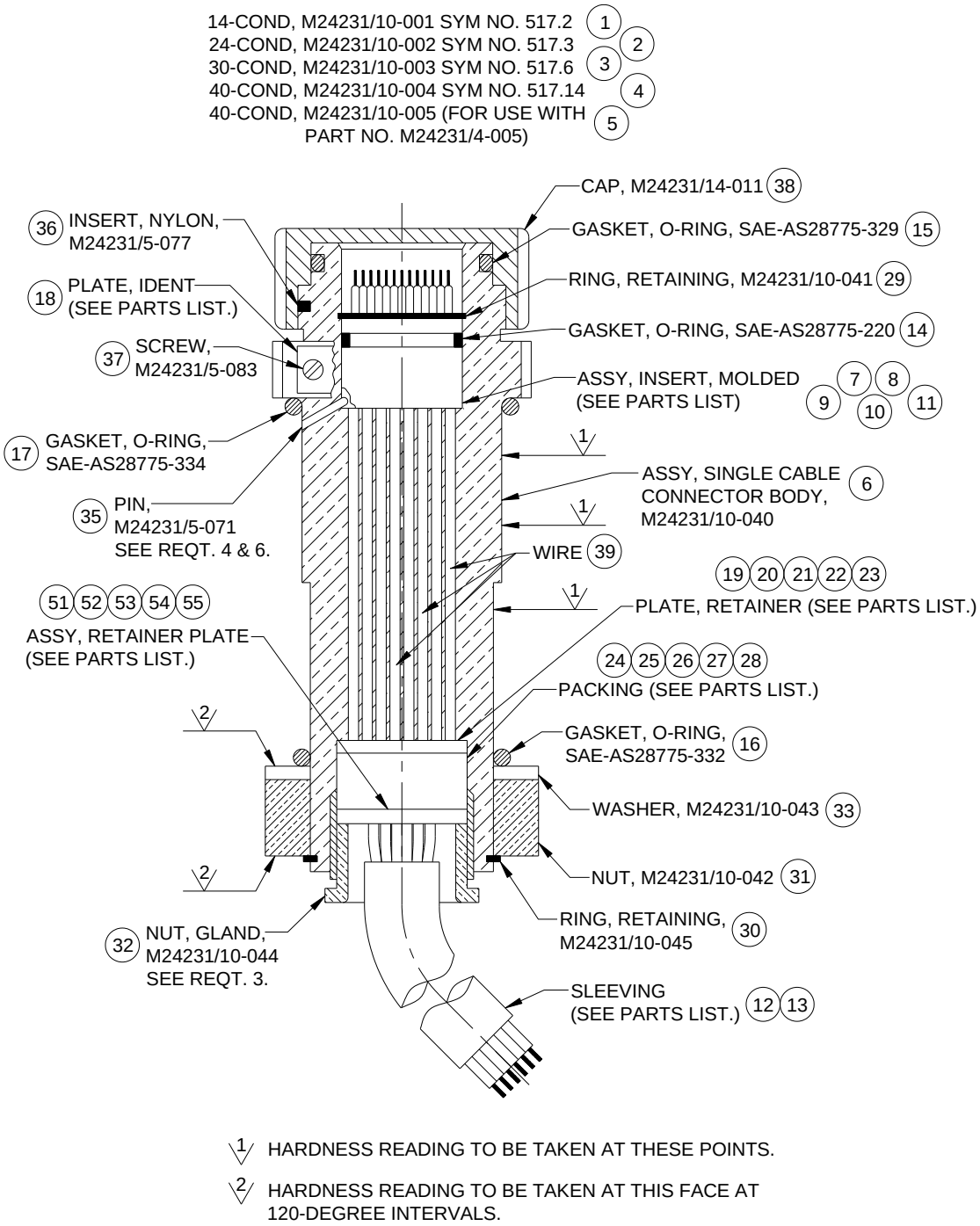


FIGURE 1. Connector, single cable, 14-, 24-, 30-, or 40-conductor.

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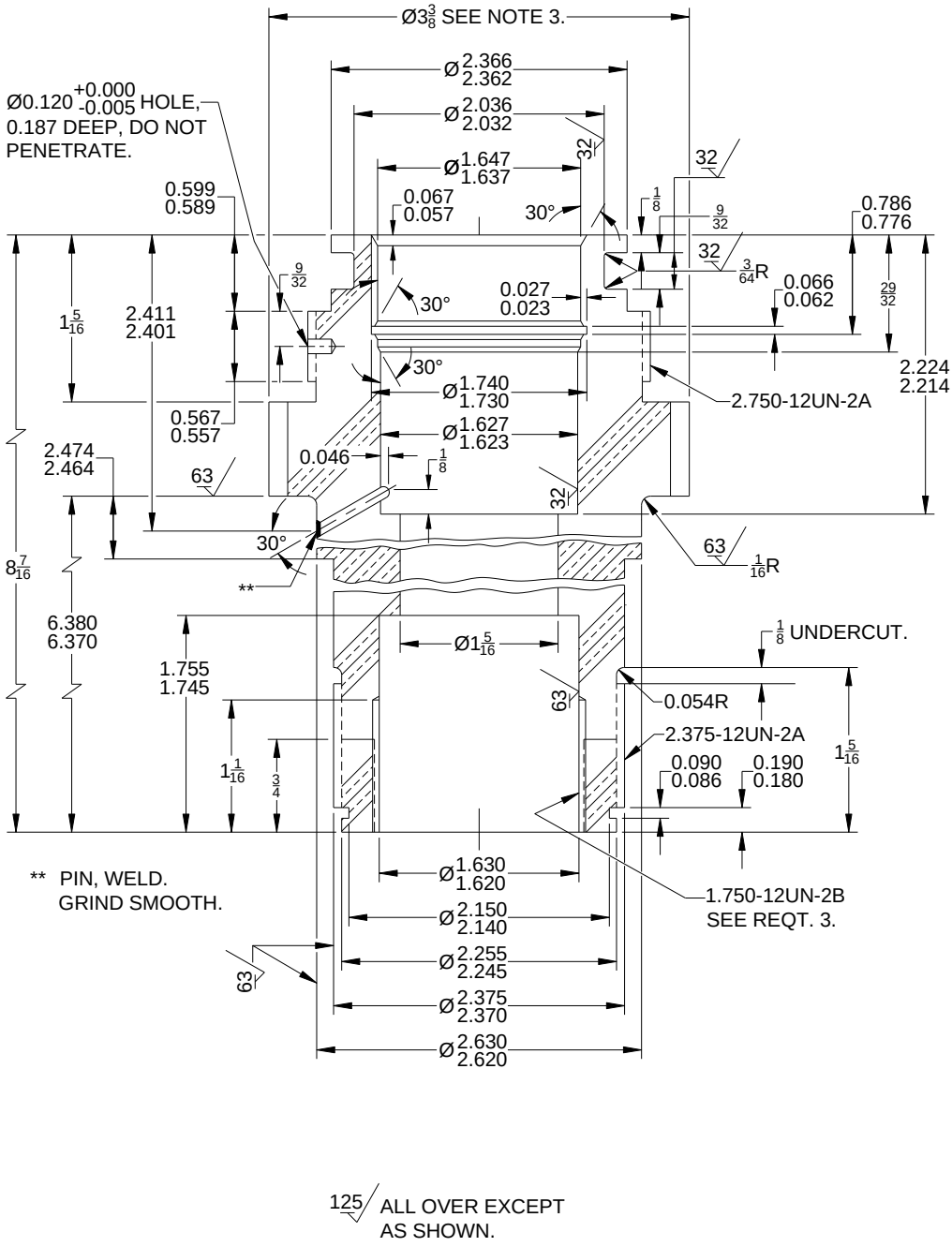


FIGURE 2. Assembly, single cable connector body – M24231/10-040. (6)

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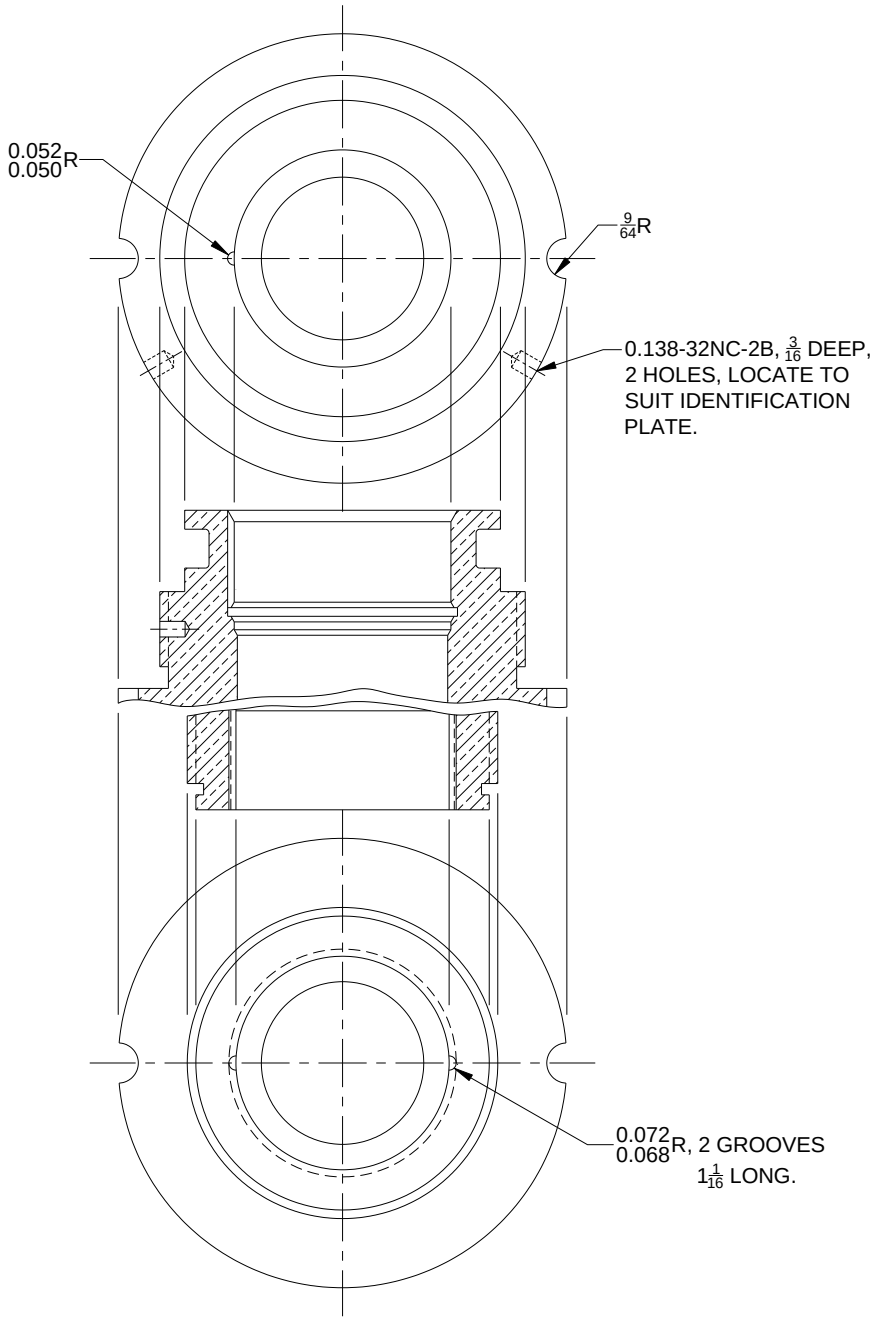


FIGURE 2. Single cable connector body – M24231/10-040 – Continued. (6)

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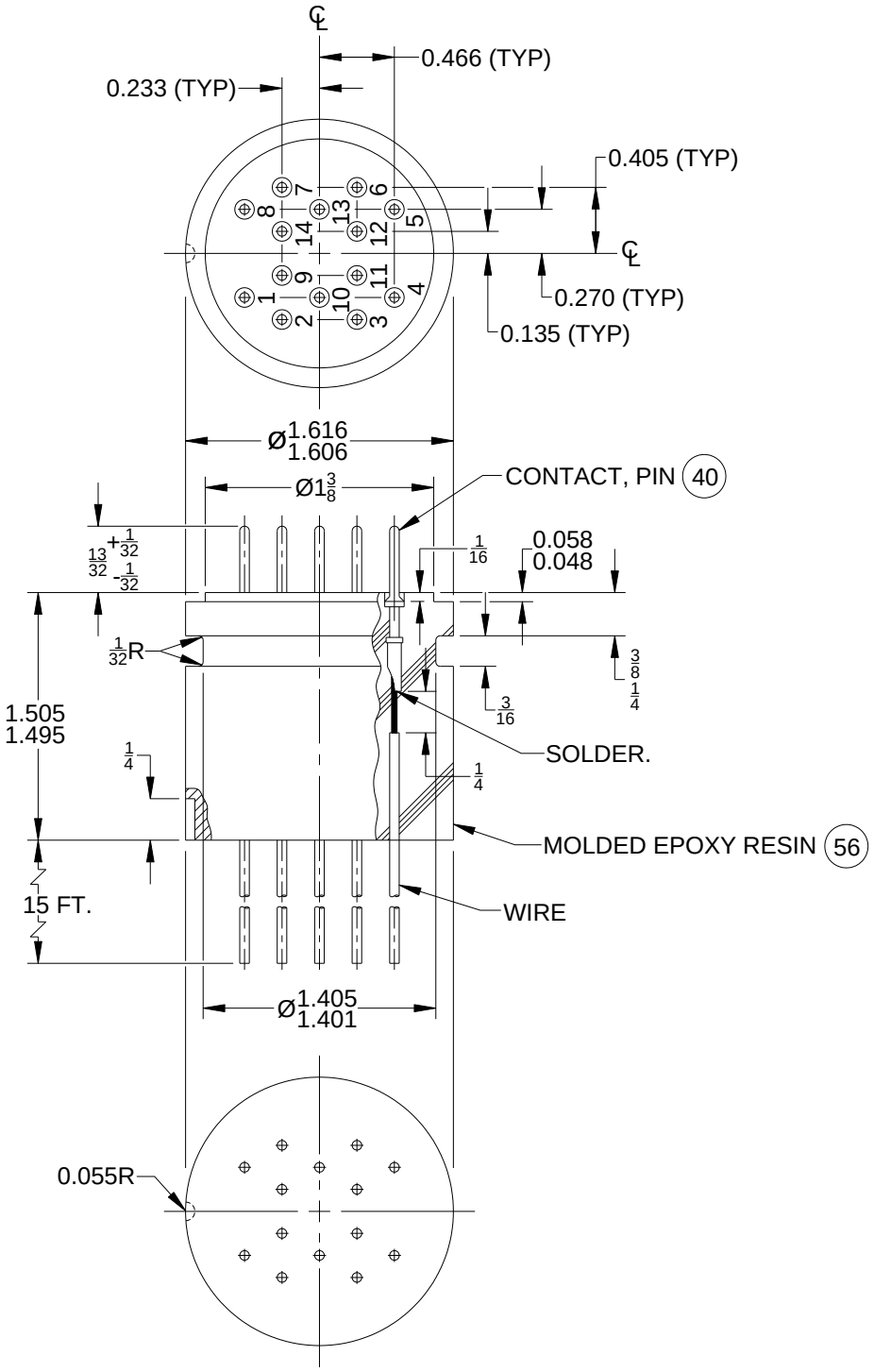


FIGURE 3. Assembly, insert, molded (14-conductor) – M24231/10-010. (7)

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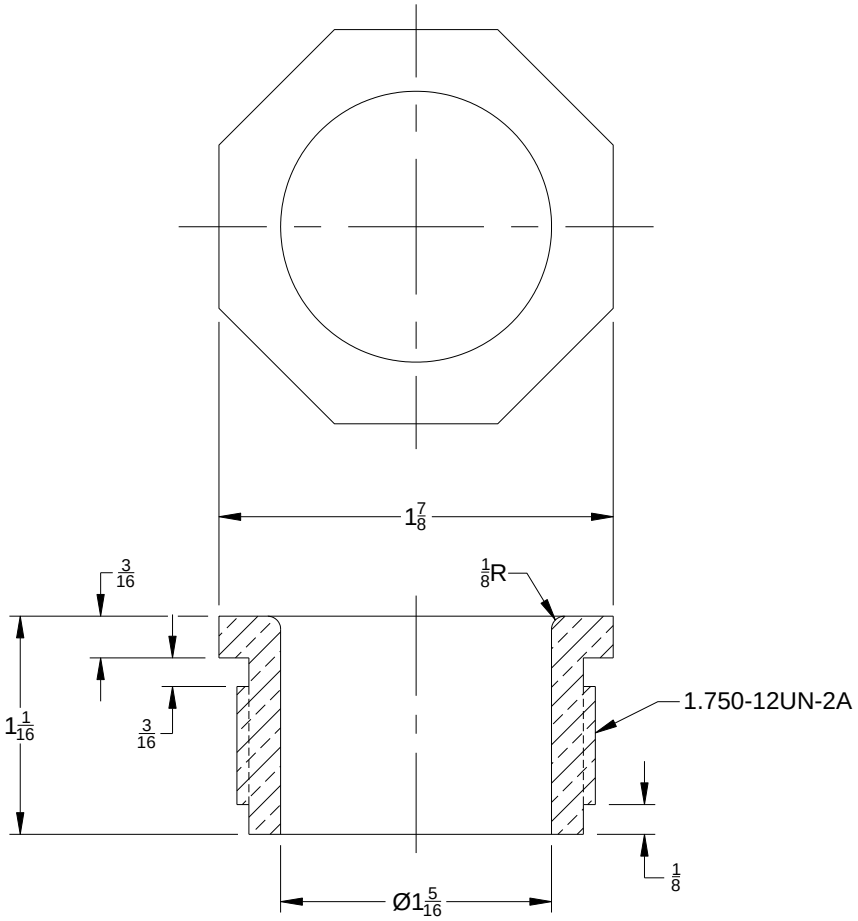
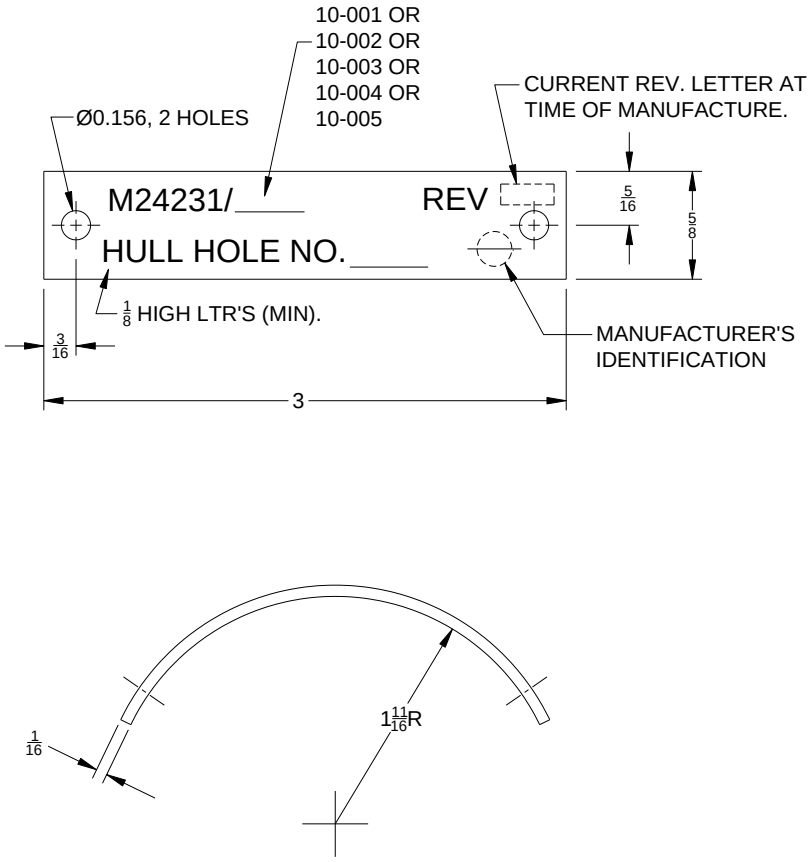


FIGURE 4. Nut, gland – M24231/10-044. (32)

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M24231/10-091 (14-COND)
M24231/10-092 (24-COND)
M24231/10-093 (30-COND)
M24231/10-094 (40-COND)
M24231/10-095 (40-COND)

FIGURE 5. Plate, identification. (18)

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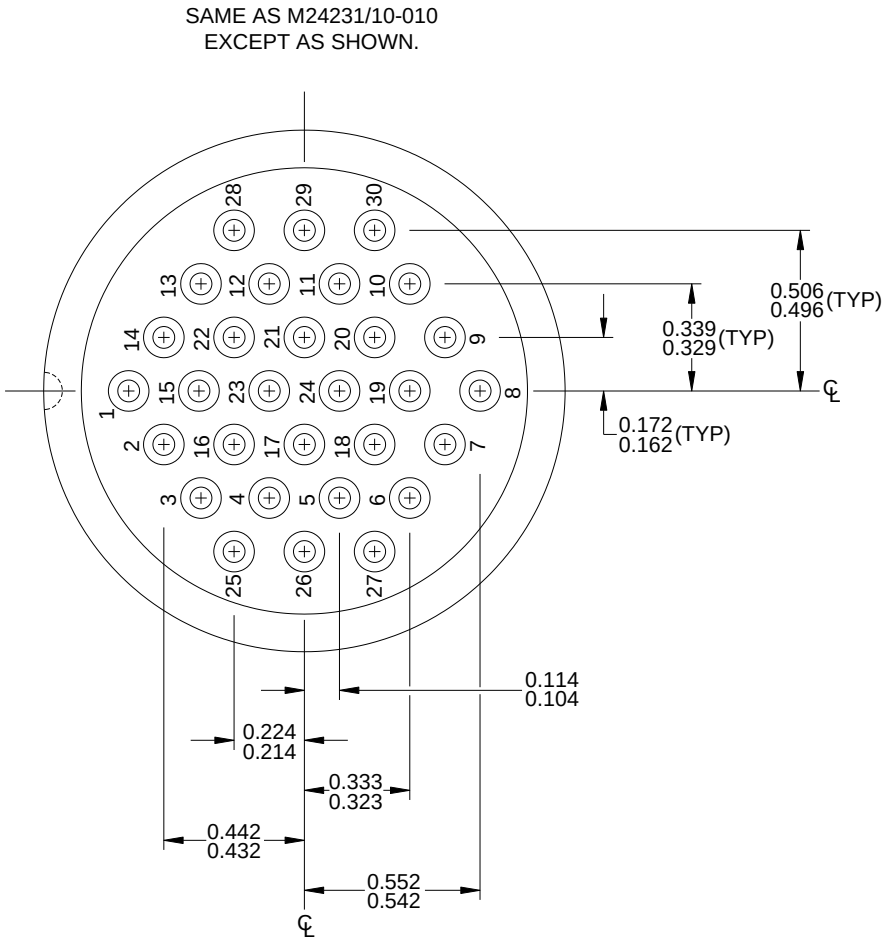


FIGURE 6. Assembly, insert, molded (30-conductor) – M24231/10-030. (9)

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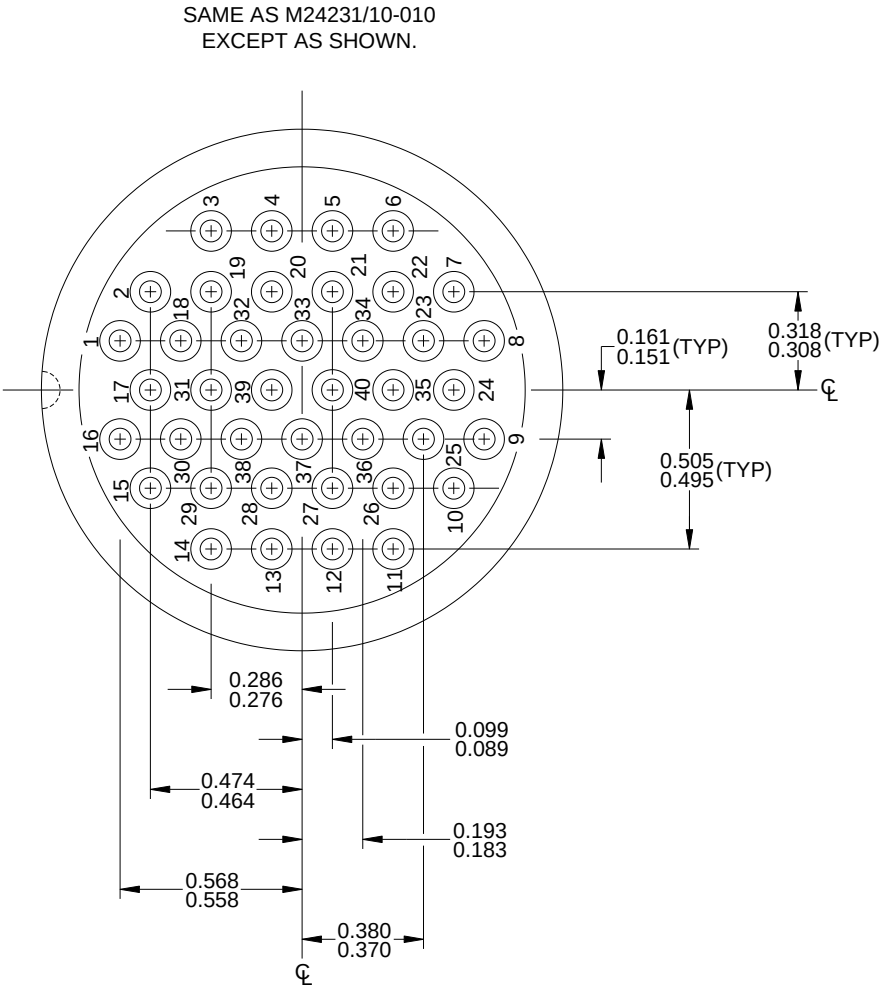


FIGURE 7. Assembly, insert, molded (40-conductor) – M24231/10-035. (10)

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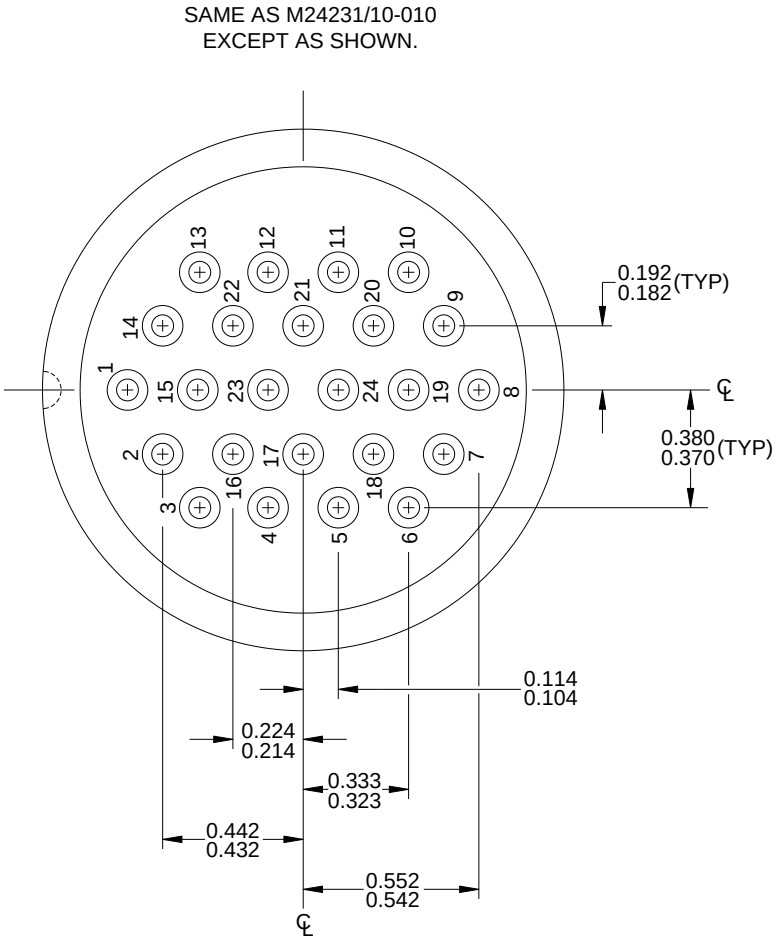


FIGURE 8. Assembly, insert, molded (24-conductor) – M24231/10-020. (8)

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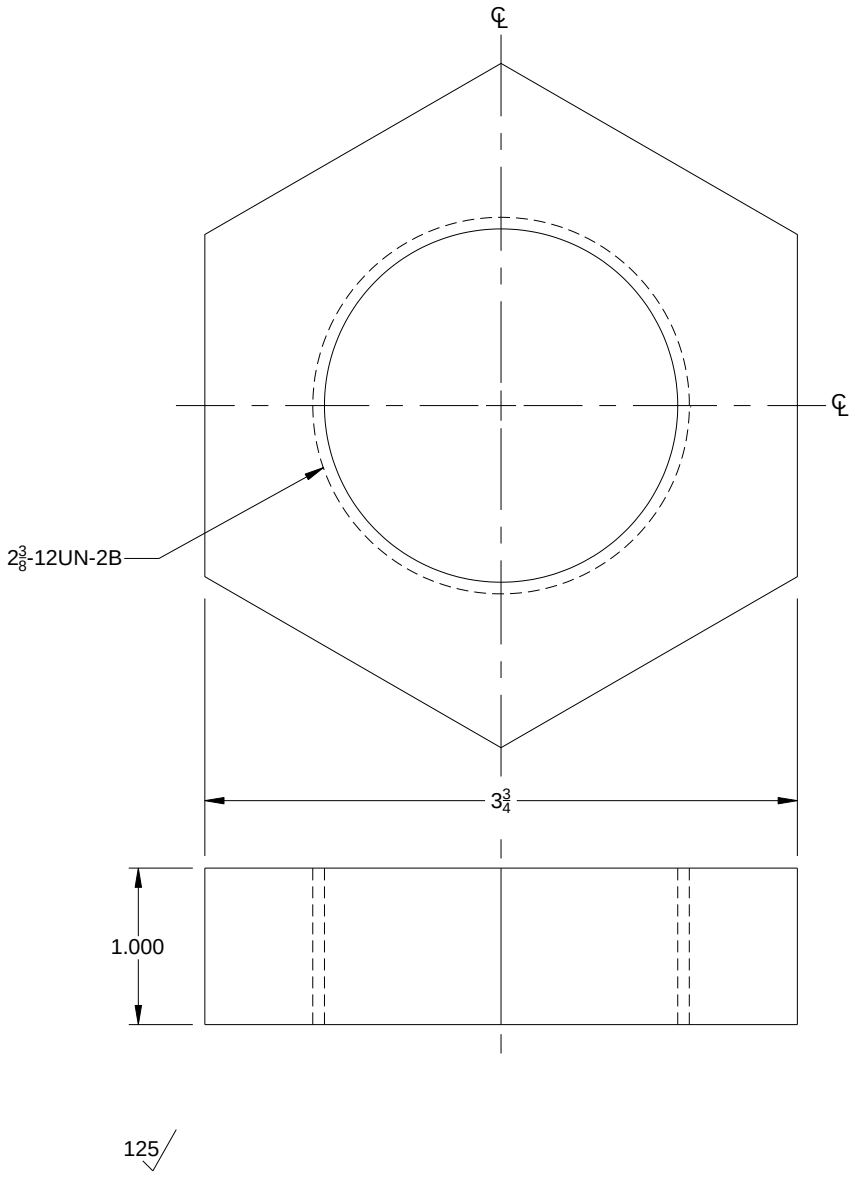


FIGURE 9. Nut – M24231/10-042. (31)

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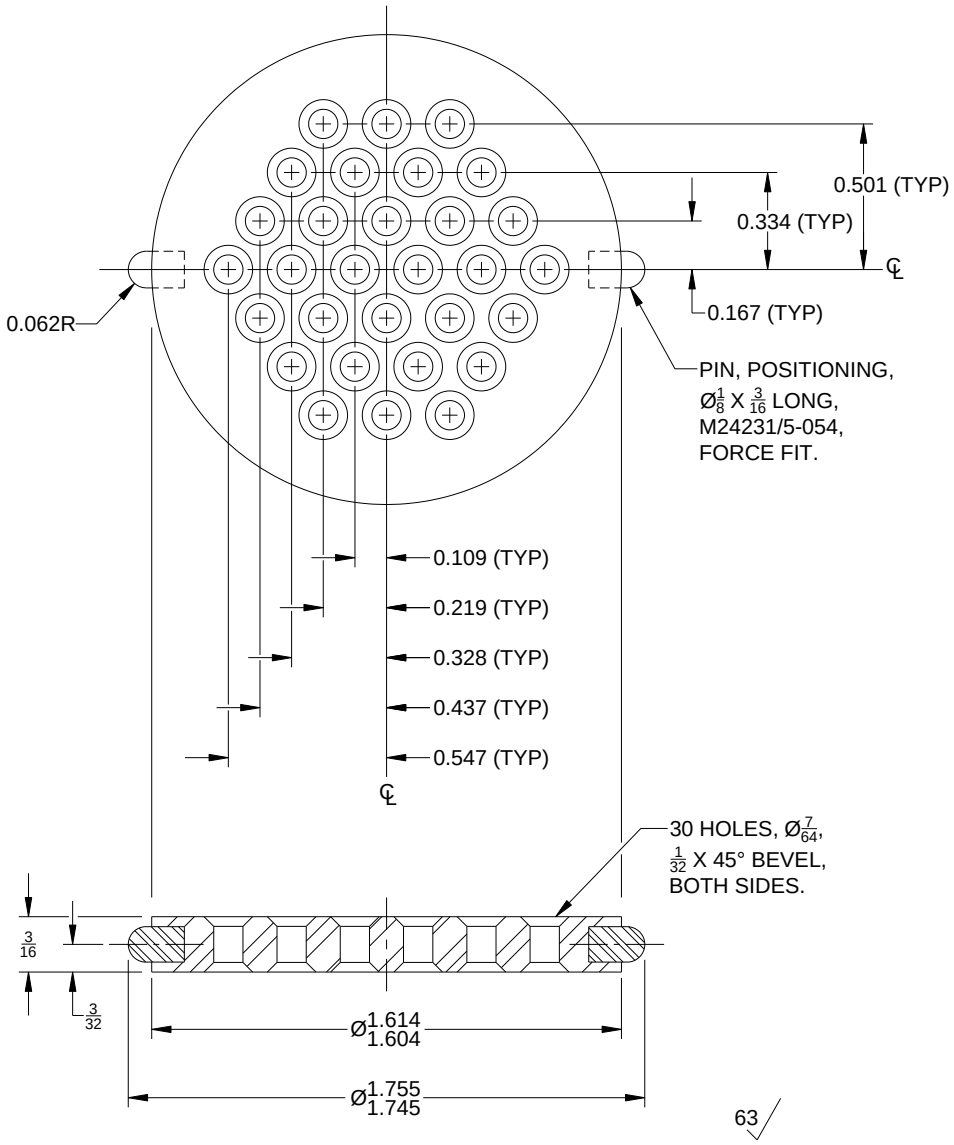


FIGURE 10. Assembly, retainer plate – M24231/10-080. (53)

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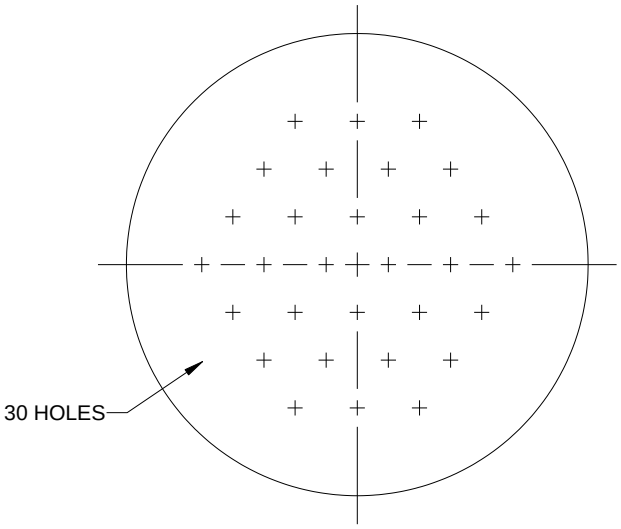


FIGURE 11. Plate, retainer (same as M24231/10-080 except no positioning pins required) – M24231/10-081. (21)

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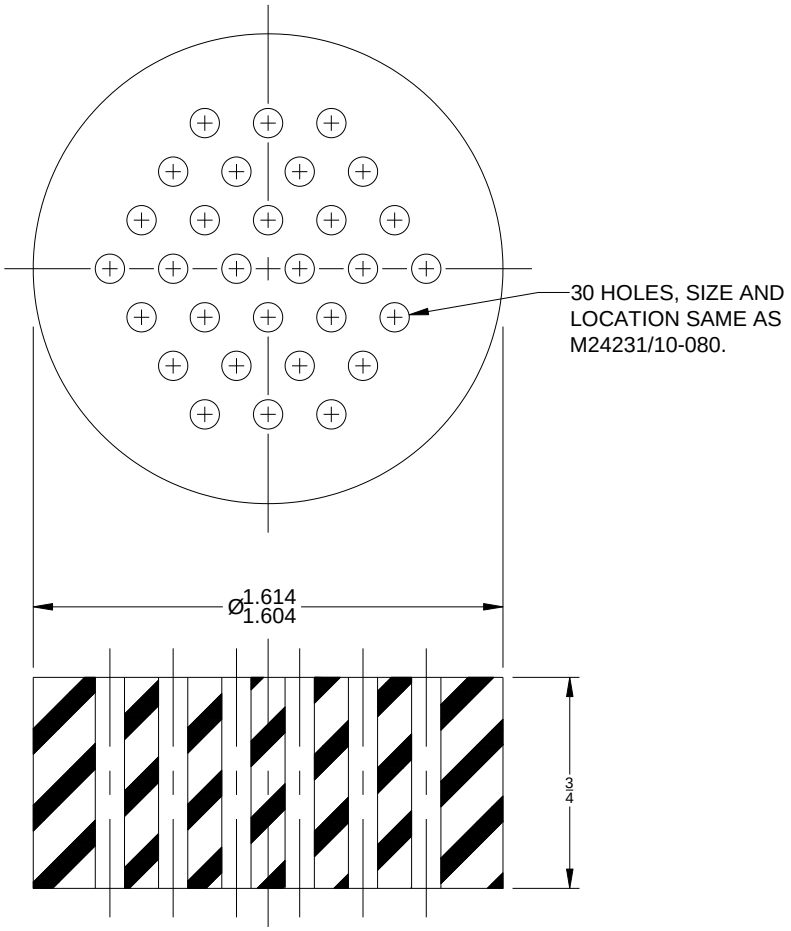


FIGURE 12. Packing – M24231/10-082. (26)

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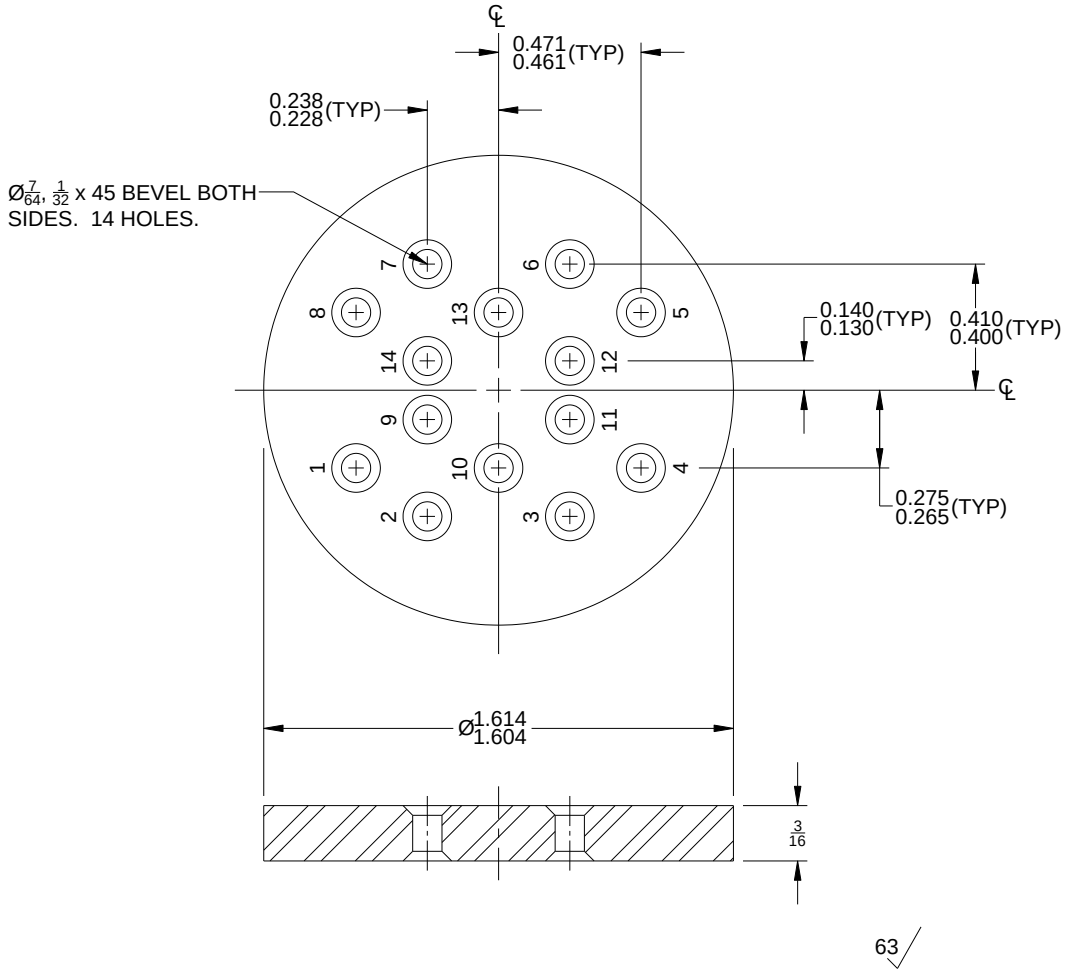


FIGURE 13. Plate, retainer – M24231/10-061. (19)

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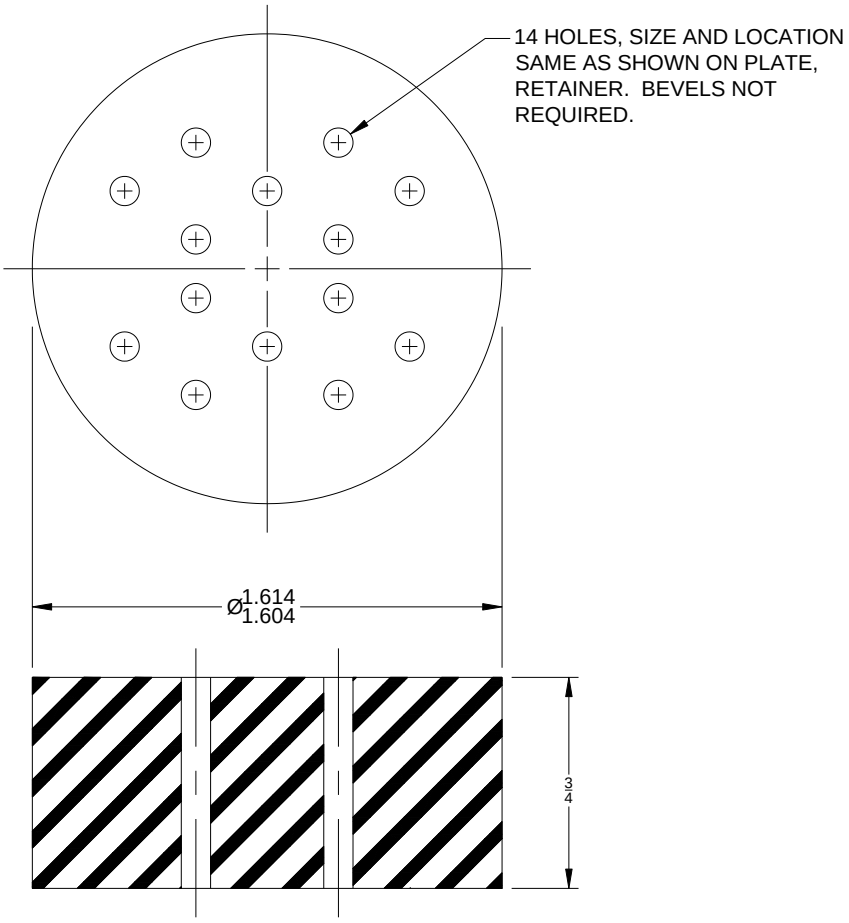


FIGURE 14. Packing – M24231/10-062. (24)

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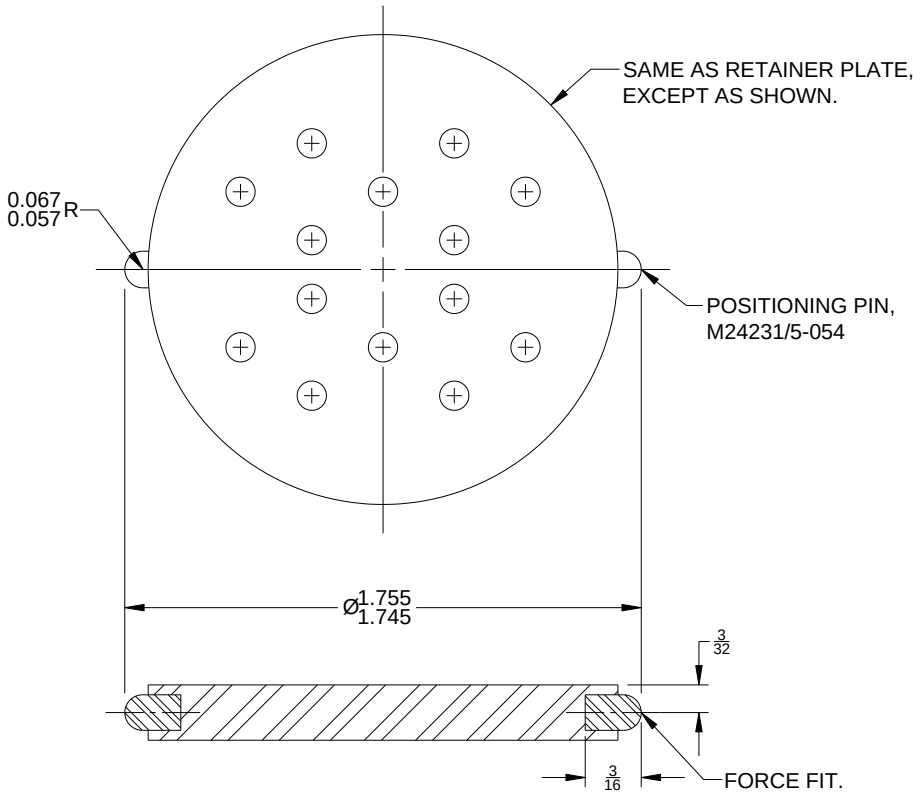


FIGURE 15. Assembly, retainer plate – M24231/10-060. (51)

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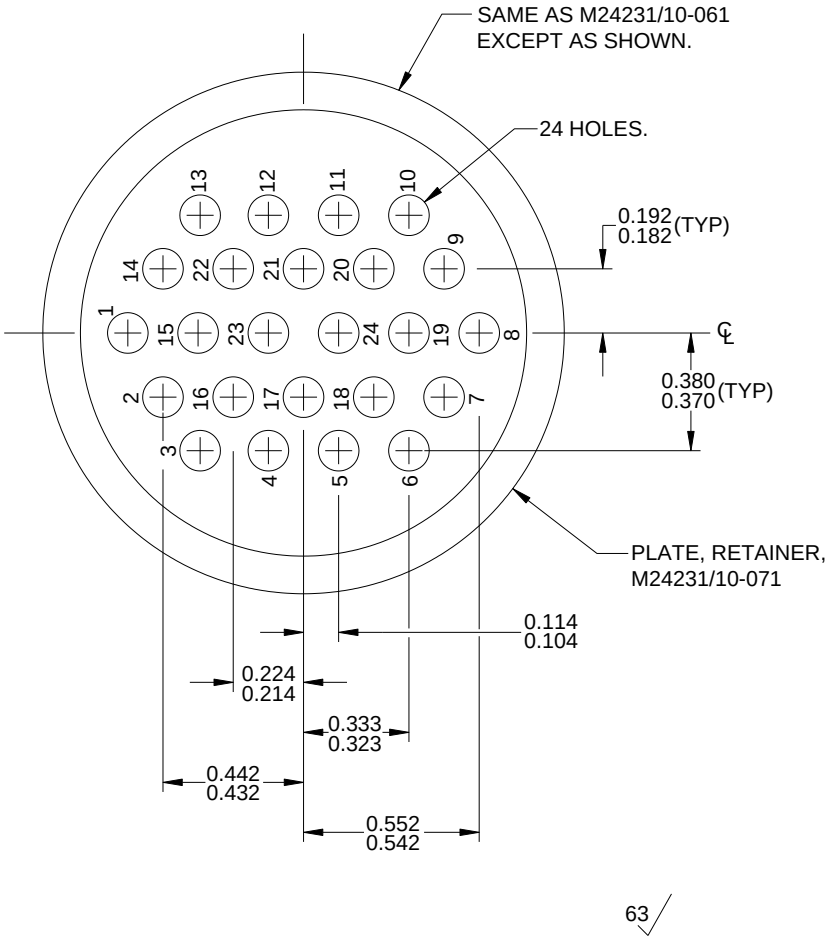


FIGURE 16. Plate, retainer – M24231/10-071. (20)

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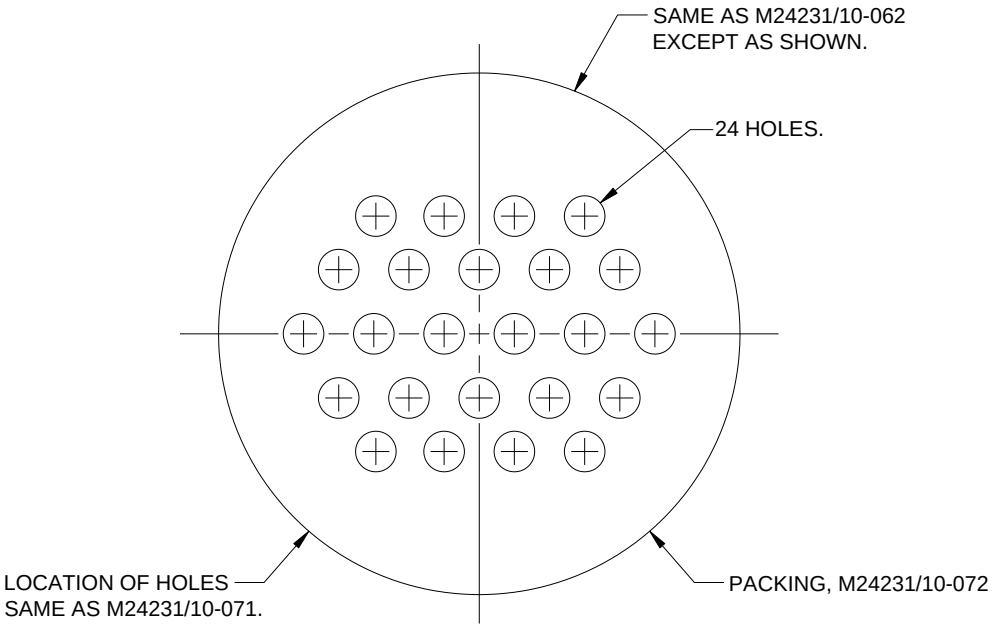


FIGURE 17. Packing – M24231/10-072. (25)

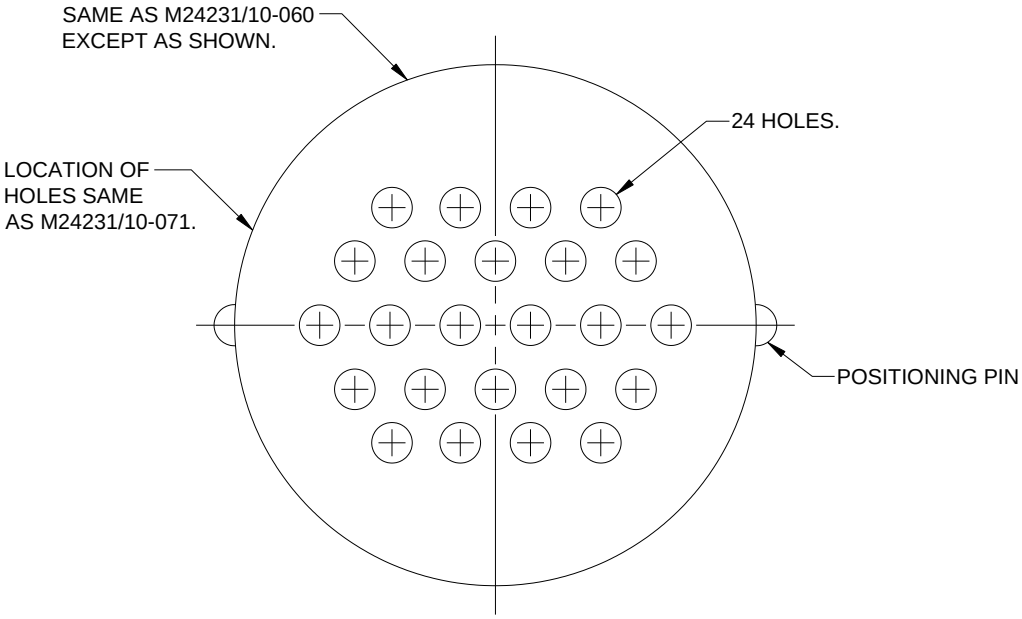


FIGURE 18. Assembly, retainer plate – M24231/10-070. (52)

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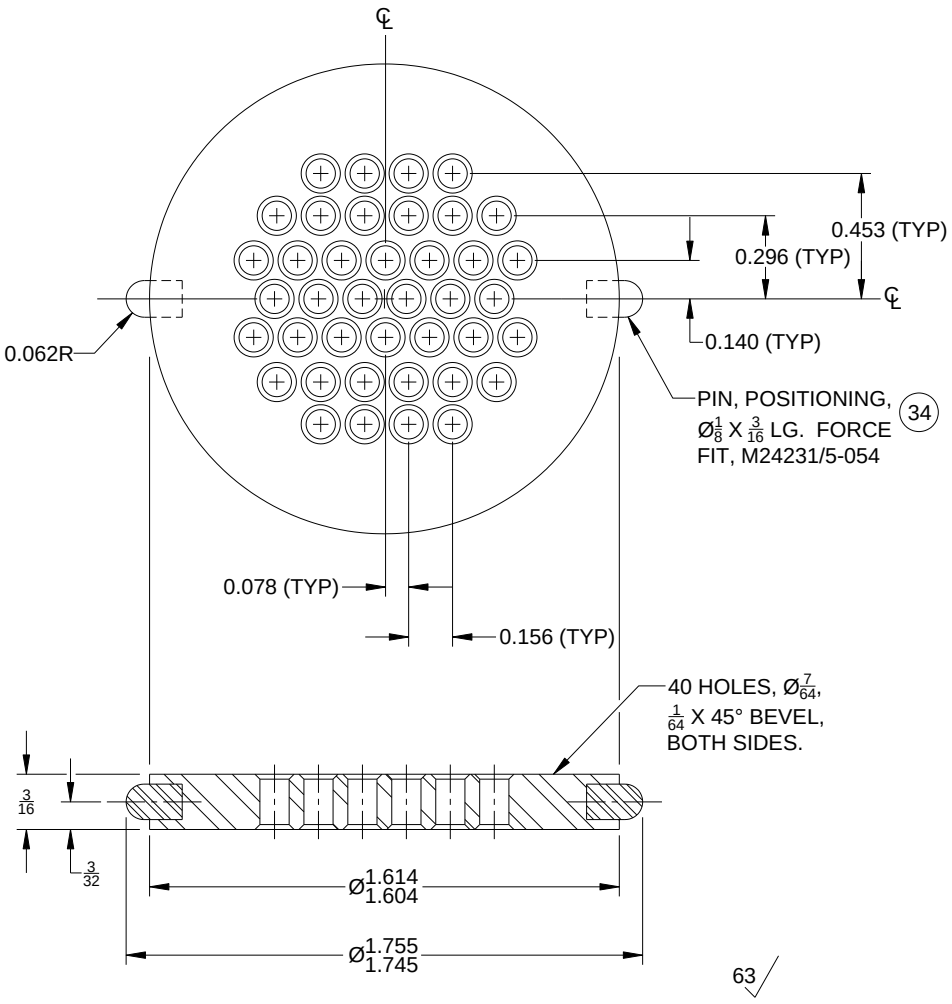


FIGURE 19. Assembly, plate, retainer – M24231/10-077. (54)

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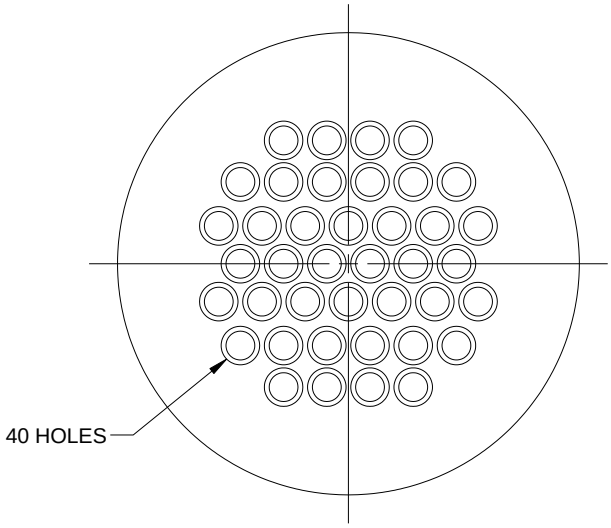


FIGURE 20. Plate, retainer (same as M24231/10-077 except no positioning pins required) – M24231/10-078. 22

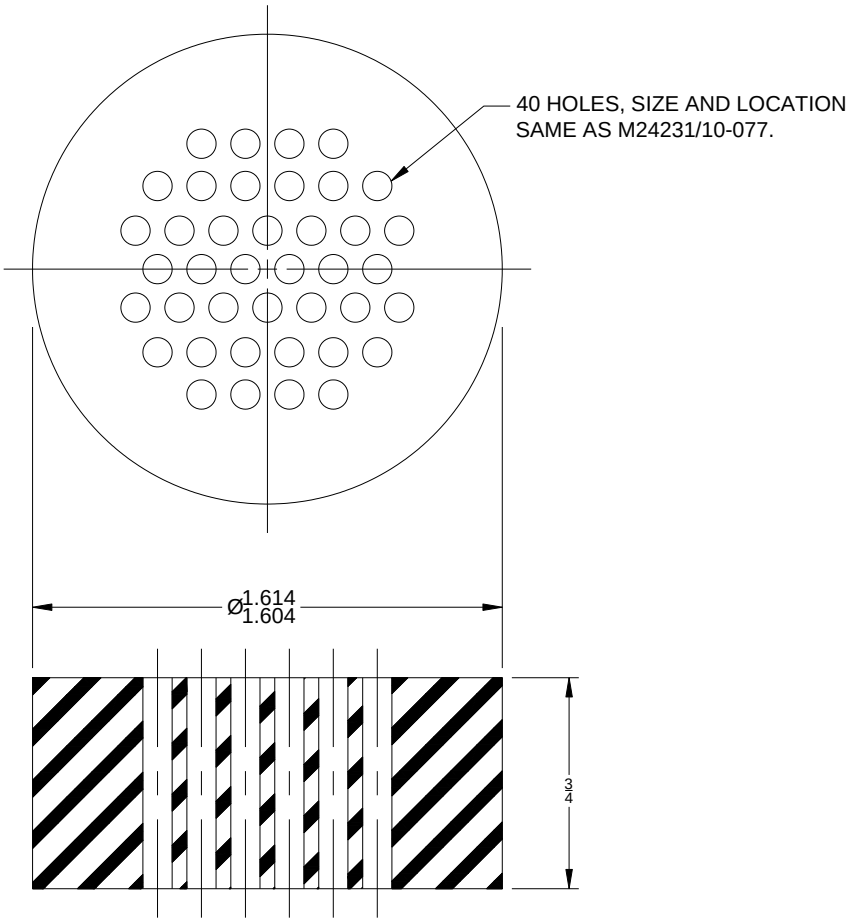


FIGURE 21. Packing – M24231/10-079. 27

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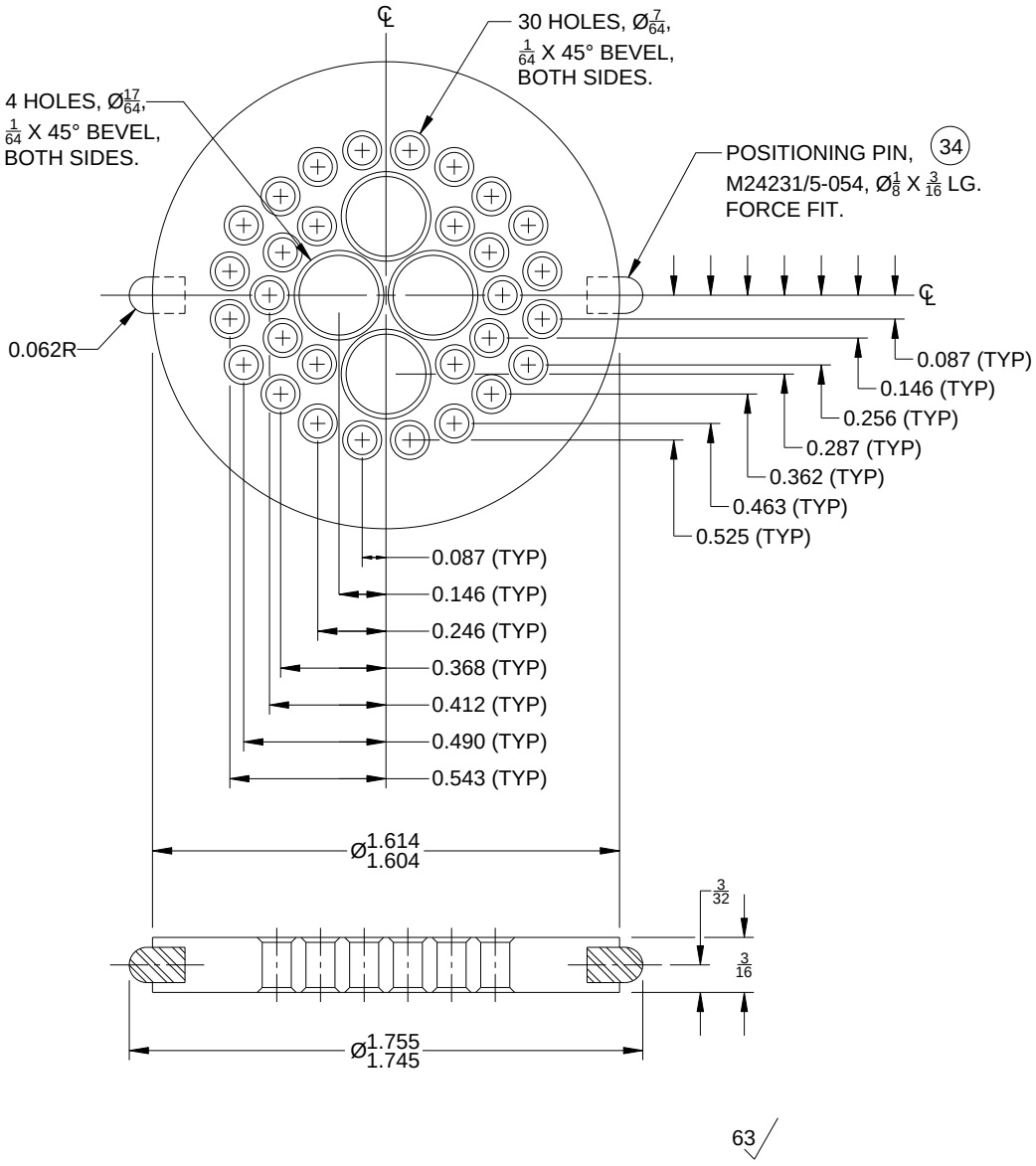


FIGURE 22. Assembly, plate, retainer – M24231/10-074. 55

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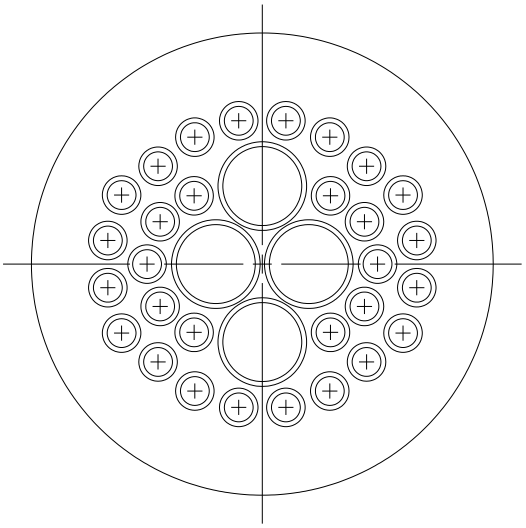


FIGURE 23. Plate, retainer (same as M24231/10-074 except no positioning pins required) – M24231/10-075. (23)

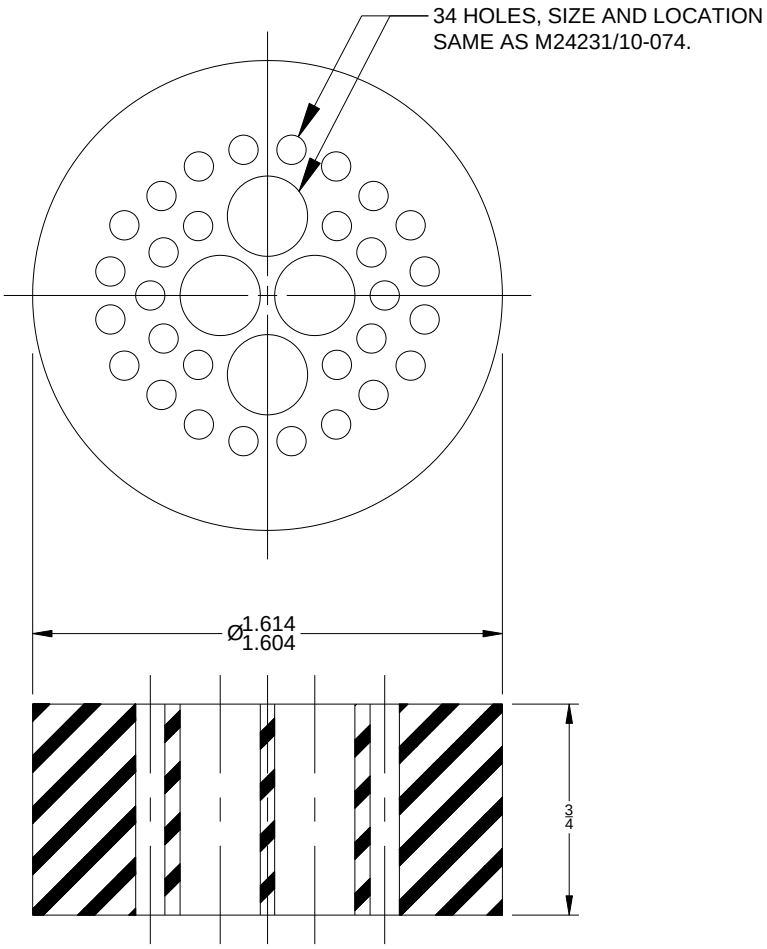


FIGURE 24. Packing – M24231/10-076. (28)

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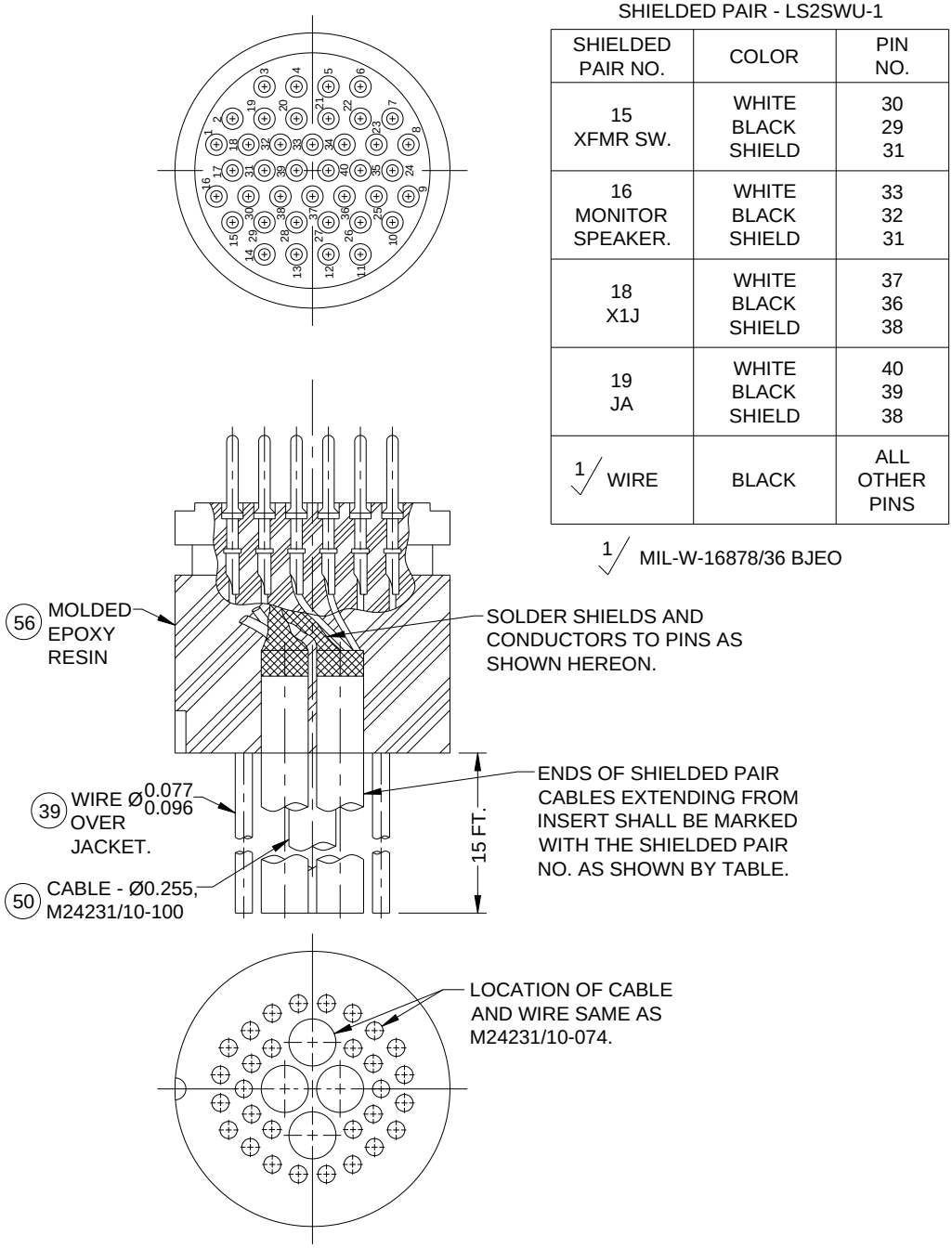


FIGURE 25. Assembly, insert, molded (40-conductor) - M24231/10-036 (same as M24231/10-035 except as shown). (11)

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TABLE I. Parts list.

Item no. 1/	Description 2/	Quantity for M24231/10- 3/					Material/specification	Remarks
		001	002	003	004	005		
1	Connector assembly, single cable, 14-conductor, symbol no. 517.2	1					-----	See figure 1.
2	Connector assembly, single cable, 24-conductor, symbol no. 517.3		1				-----	See figure 1.
3	Connector assembly, single cable, 30-conductor, symbol no. 517.6			1			-----	See figure 1.
4	Connector assembly, single cable, 40-conductor, symbol no. 517.14				1		-----	See figure 1.
5	Connector assembly, single cable, 40-conductor					1	-----	See figure 1.
6	Connector body, single cable	1	1	1	1	1	Ni-Cu-Al, QQ-N-286	See figure 2 and continuation of figure 2 and requirement 4.
7	Insert assembly, molded 14-conductor	1					See items 39, 40, and 56	See figure 3.
8	Insert assembly, molded 24-conductor		1				See items 39, 40, and 56	See figure 3.
9	Insert assembly, molded 30-conductor			1			See items 39, 40, and 56	See figure 3.
10	Insert assembly, molded 40-conductor				1		See items 39, 40, and 56	See figure 3.
11	Insert assembly, molded 40-conductor					1	See items 39, 40, and 56	See figure 3.
12	Sleeving, 0.680 id by 0.040 thick	15 ft					Type F, Form U, Category 1, color clear, MIL-I-631	See figure 1.
13	Sleeving, 0.890 id by 0.050 thick		15 ft	15 ft	15 ft	15 ft	See item 12	See figure 1.
14	Gasket, O-ring	1	1	1	1	1	SAE-AS28775-220 4/	Lubricate with silicone compound, SAE-AS8660; see figure 1.
15	Gasket, O-ring	1	1	1	1	1	SAE-AS28775-329 4/	See item 14.
16	Gasket, O-ring	1	1	1	1	1	SAE-AS28775-332 4/	See item 14.
17	Gasket, O-ring	1	1	1	1	1	SAE-AS28775-334 4/	See item 14.
18	Plate identification	1	1	1	1	1	CRES or Al, MIL-DTL-15024	See figure 5.

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TABLE I. Parts list – Continued.

Item no. 1/	Description 2/	Quantity for M24231/10- 3/					Material/specification	Remarks
		001	002	003	004	005		
19	Plate, retainer	1					Ni-Cu, Class A, QQ-N-281	See figure 13 and requirement 5.
20	Plate, retainer		1				See item 19.	See figure 16 and requirement 5.
21	Plate, retainer			1			See item 19.	See figure 11 and requirement 5.
22	Plate, retainer				1		See item 19.	See figure 20 and requirement 5.
23	Plate, retainer					1	See item 19.	See figure 23 and requirement 5.
24	Packing, Class A, durometer 55±5	1					Rubber, ZZ-R-768	See figure 14.
25	Packing, Class A, durometer 55±5		1				Rubber, ZZ-R-768	See figure 17.
26	Packing, Class A, durometer 55±5			1			Rubber, ZZ-R-768	See figure 12.
27	Packing, Class A, durometer 55±5				1		Rubber, ZZ-R-768	See figure 21.
28	Packing, Class A, durometer 55±5					1	Rubber, ZZ-R-768	See figure 24.
29	Ring, retaining	1	1	1	1	1	SPIROLOX RR-165-S, or equal	See figure 1.
30	Ring, retaining	1	1	1	1	1	TRUARC 5100-225-W, or equal	See figure 1.
31	Nut	1	1	1	1	1	Ni-Cu-Al, QQ-N-286	See figure 9.
32	Nut, gland	1	1	1	1	1	Ni-Al-BRZ, ASTM B150/B150M	See figure 4 and requirement 2.
33	Washer, 4⅜ od by 2.385 id by 0.162 thick	1	1	1	1	1	Ni-Cu-Al, QQ-N-286	See figure 1; finish shall be 63 microinches.
34	Pin, positioning	2	2	2	2	2	M24231/5-054	See figure 10, figure 15, figure 18, figure 19, and figure 22.
35	Pin, guide	1	1	1	1	1	M24231/5-071	See requirements 4 and 6.
36	Insert	1	1	1	1	1	M24231/5-077	See figure 1.
37	Screw	1	1	1	1	1	M24231/5-083	See figure 1.
38	Cap, pressure-proof	1	1	1	1	1	M24231/14-011	See figure 1.
39	Wire, 15 feet	As required					MIL-DTL-16878/36 BJE0	See figure 1.

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TABLE I. Parts list – Continued.

Item no. 1/	Description 2/	Quantity for M24231/10- 3/					Material/specification	Remarks
		001	002	003	004	005		
40	Pin, contact	14	24	30	40	40	MIL-DTL-24231	See figure 3.
50	Cable, Type LS2SWU-1, 15 ft						MIL-DTL-24643/33	See figure 25.
51	Assembly, retainer plate	1					Ni-Cu, Class A, QQ-N-281	See figure 15 and requirement 5.
52	Assembly, retainer plate		1				Ni-Cu, Class A, QQ-N-281	See figure 18 and requirement 5.
53	Assembly, retainer plate			1			Ni-Cu, Class A, QQ-N-281	See figure 10 and requirement 5.
54	Assembly, retainer plate				1		Ni-Cu, Class A, QQ-N-281	See figure 19 and requirement 5.
55	Assembly, retainer plate					1	Ni-Cu, Class A, QQ-N-281	See figure 22 and requirement 5.
56	Epoxy, resin, molded	As required					Cast epoxy resin, compound 5/	See figure 3.

NOTES:

1/ Item numbers are the same as encircled numbers shown on figures.

2/ See applicable figure or table listed in remarks column for part numbers and dimensions.

3/ Quantity is for one complete assembly of each size.

4/ Nitrile (Buna N); Parker Hannifin Corporation (part number N0756-75), O-ring Division, 2360 Palumbo Drive, Lexington, KY 40509, CAGE 02697, or equal.

5/ Arco Set 200, M.M. Reif and Co., Ivyland, PA PR610-TCF (Chockfast Orange), ITW Philadelphia Resins Corp., Montgomeryville, PA, or equal.

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CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Preparing activity:
Navy – SH
(Project 5935-2011-059)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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